

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

Code

```
string = input("Enter a string: ")

if string.endswith("ab"):

    print("Accepted")

else:

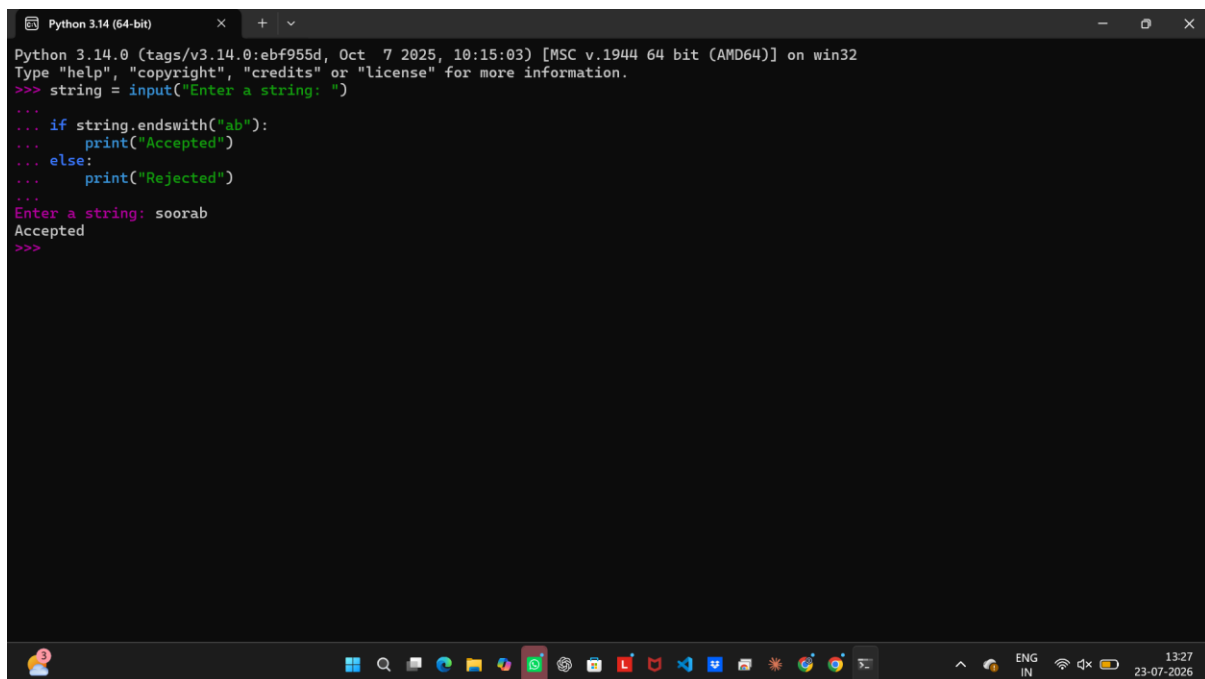
    print("Rejected")
```

Sample Input

soorab

Sample Output

Accepted

A screenshot of a Windows terminal window titled "Python 3.14 (64-bit)". The terminal shows the execution of a Python script. The script prompts the user to "Enter a string: " and the user enters "soorab". The script then checks if the string ends with "ab" and prints "Accepted". The terminal output is as follows:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> string = input("Enter a string: ")
...
... if string.endswith("ab"):
...     print("Accepted")
... else:
...     print("Rejected")
...
Enter a string: soorab
Accepted
>>>
```

The terminal window has a dark background with light-colored text. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 13:27 on 23-07-2026.